

The Pacific Report

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FRONT PAGE

Jupiter's Lightning May Have the Force of Nuclear Weapons

*Editor*David · *Slashdot*

How powerful is Jupiter's lightning? Thick clouds cover the view, notes Science magazine. But using an instrument on NASA's Juno spacecraft (orbiting Jupiter for the past decade), researchers determined Jupiter's lightning bolts are 100 to 10,000 times more energetic than earth's:

A single bolt of lightning on Earth releases about 1 billion joules of energy. That means the most extreme bolts of jovian lightning carry 10 trillion joules of energy, equivalent to 2400 tons of TNT, or one-sixth the power of the atomic bomb dropped on Hiroshima, Japan. Based on the rates of flashes seen by Juno, storms on this tempestuous world can unleash the force of multiple nuclear weapons every minute...

The four storms Juno studied were monstrous, says Michael Wong, a planetary scientist at the University of California, Berkeley and one of the study's authors. There were three flashes per second on average, often emerging from the hearts of storms that are 3000 kilometers across, longer than the distance from New York City to Denver. The researchers used the Hubble Space Telescope (and photographs from Juno's camera) to track Jupiter's storms with such precision that their radiometer could then pick out individual lightning flashes, according to the article.

"It's just a massive ball of gas. It makes sense that there's very energetic lightning happening," says Daniel Mitchard, a lightning physicist at Cardiff University who wasn't involved with the new study. But confirming such suspicions "is exciting," he says, because lightning plays an important role in forging complex chemistry — including the sort that primordial life is built on.

Thanks to Slashdot reader sciencehabit for sharing the article.

Read more of this story at Slashdot.

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EditorDavid

FEATURES

Step-by-Step Guide to Apply for a Business Loan

Jessica Clark · Small Business Trends

Applying for a business loan can seem intimidating, but breaking it down into manageable steps simplifies the process. Start by comprehending the types of loans available and evaluating your specific funding needs. Reviewing your credit score and financial health is essential, as these factors influence your eligibility. Gathering the right documentation and completing the application accurately can greatly improve your chances of approval. Ready to learn what comes next after you submit your application?

Identify your business needs and determine the exact funding amount required for expansion, operations, or equipment purchases.

Assess your eligibility by reviewing your credit score, business longevity, and financial health, ensuring you meet lender criteria.

Compare lenders and loan options, evaluating interest rates, repayment terms, and additional fees to find the best fit.

Gather necessary documentation, including financial statements, personal information, and a business plan, to support your application.

Complete and submit the application accurately, then review the loan agreement for terms, fees, and repayment schedules before signing.

Comprehending business loans is crucial for any entrepreneur looking to secure funding for their ventures. A business loan is an agreement between you and a lender, providing capital that needs repayment with interest. You can use these loans for various purposes, like covering operating expenses, funding expansion, or purchasing equipment.

Business loans come in two main types: secured, backed by collateral, and unsecured, based on your creditworthiness. The approval process often hinges on your credit score, business longevity, and overall financial stability, including cash flow and revenue.

Before applying, verify your business registration in the US is complete, as this can impact your eligibility. Grasping the terms, such as repayment periods that can range from months to years, will help you make informed decisions.

Knowing the specifics of different loan types can likewise guide you in finding the best option for your business needs.

Types of Business Loans

When exploring types of business loans, it's essential to recognize the variety of options available to meet your financial needs.

Business lines of credit provide flexible, revolving access to cash, perfect for daily expenses. If you're looking to manage expenses during building credit, consider business credit cards designed particularly for that purpose.

Traditional business term loans offer a lump sum that you repay in fixed increments, making them suitable for larger investments, like expansions. For small businesses, SBA small business loans are government-backed, offering favorable terms and lower interest rates.

On the other hand, merchant cash advances give you upfront capital in exchange for a percentage of future sales. Although this can be a quick funding option, it often comes at a higher cost.

Comprehending these various types of business loans will help you make an informed decision when seeking financial assistance for your business.

How can you effectively assess your business needs when considering a loan? Start by identifying the specific purpose of the loan. Are you looking to fund expansion, purchase equipment, or cover operational expenses?

Next, determine the amount of funding required by analyzing projected costs to ascertain it aligns with your financial goals. Conduct a cash flow assessment to confirm your business can manage loan repayments comfortably, ideally maintaining income at least 1.25 times total expenses.

Evaluate your current financial health, including annual revenue and credit score.

Consider the type of loan that best suits your situation, whether it's a traditional term loan for major purchases or a line of credit for daily needs.

Be clear about your business goals to guide your decision-making.

Determining Eligibility for a Business Loan

Grasping your business needs is just the first step in the loan application process; determining your eligibility is equally important. To qualify for a business loan, most lenders prefer a minimum credit score of 690.

If you're figuring out how to start a business in America, bear in mind that your business should ideally be operational for at least 1 to 2 years, though some online lenders may accept those with just 3 to 12 months in business.

Furthermore, you'll need to meet annual revenue requirements, which can range from \$50,000 to \$250,000, depending on the lender and loan type. Demonstrating a steady cash flow, ideally 1.25 times your total expenses, is vital for proving your ability to repay the loan.

Finally, keep in mind that different lenders have varying eligibility criteria influenced by your industry and the presence of collateral for secured loans.

HP Launches Next-Gen LaserJet Printers with Quantum-Resistant Security and AI Tools

David Wilson · Small Business Trends

HP has unveiled two new printer series designed to help small and medium-sized businesses (SMBs) effectively bridge the gap between paper and digital workflows while enhancing security. The HP LaserJet Pro 4000/4100 Series and the HP LaserJet Enterprise 5000/6000 Series promise to address the critical needs of businesses looking to streamline operations and protect sensitive data amid rising cyber threats.

“Organizations are under pressure to digitize how work gets done while keeping their environments secure,” said Anneliese Olson, President of Imaging, Printing, and Solutions at HP Inc. This statement reflects the growing urgency among SMBs; a recent report found that 78% face challenges managing hardware and supplies, while 80% plan to adopt or expand print-related software for efficiency.

The LaserJet Pro 4000/4100 Series, tailored for smaller teams, aims to deliver high-quality monochrome printing in a compact design. Its two-sided printing is touted as the fastest in its class, making it an ideal choice for businesses with limited IT resources. The easy setup is complemented by efficient connectivity options, including Wi-Fi 6, which purportedly self-resets for uninterrupted service.

While the focus on security is crucial, HP claims the Pro series is the world’s first small business printer to feature quantum-resistant protection. This addition is timely, as cybersecurity experts predict an alarming 34% likelihood of quantum computer attacks by 2034, emphasizing the importance of advanced defenses for SMBs.

The series offers a reported 25% reduction in ongoing print costs compared to previous models, facilitated by HP’s innovative TerraJet Toner technology. Environmentally conscious businesses will benefit from energy-efficient designs that showcase certifications like EPEAT Gold, ENERGY STAR®, and Blue Angel.

The Enterprise 5000/6000 Series targets organizations requiring high-speed, dependable performance. Capable of scanning at 200 images per minute (ipm) and printing up to 57 pages per minute (ppm), it supports demanding teams that rely heavily on efficient printing and document digitization. Its built-in smart workflow solutions leverage HP’s AI capabilities to enhance document formatting, automate routine tasks, and streamline operations.

Security remains a top concern with this enterprise-grade series. It is powered by HP Wolf Enterprise Security, which is designed to minimize cyberattack risk dramatically. Organizations adopting these printers can maintain compliance and support digital transformation efforts while being aware of potential vulnerabilities.

Both the Pro and Enterprise series promise innovative workflow solutions that simplify management without the need for extensive IT support. Features like automated

guided redaction and advanced optical character recognition (OCR) can save businesses significant processing time, cutting manual steps by as much as 50%.

While the initial adoption of these printers may require a financial commitment, the potential for long-term savings through reduced printing costs and efficient document management could prove beneficial. However, SMB owners should remain aware of possible challenges during the transition, including training staff to utilize new technologies effectively and ensuring the integration of existing systems for seamless operation.

The HP LaserJet Pro 4000/4100 Series is expected to launch in May, with pricing details forthcoming. The LaserJet Enterprise 5000/6000 Series will be available for contract sales starting in March, with broader availability in June.

In an ever-evolving business landscape, HP’s new LaserJet offerings aim to solidify the role of printing technology as a vital component of small business operations. For further details on HP’s announcements, visit the official release .

Image via Google Gemini

This article, “ HP Launches Next-Gen LaserJet Printers with Quantum-Resistant Security and AI Tools “ was first published on Small Business Trends

Weekly Update 490

Troy Hunt

Presently sponsored by: Report URI: Guarding you from rogue JavaScript! Don't get pwned; get real-time alerts & prevent breaches #SecureYourSite

A big “thank you” to everyone who helped me troubleshoot the problem with my “Print Screen” button on the new PC. Try as we all might, none of us could figure out why it refused to bind to SnagIt and instead insisted on dumping the entire collection of screens to a file on the desktop. But an especially big thanks to the follower who later emailed me with an idea that didn't work, and followed up with an idea that finally did!

So, yeah, thanks Logitech for making this a real pain in the arse ♂

How to detect Safari and iOS versions with ease in 2026

Travis Turner · Evil Martians Chronicles

Authors: Evgeniy Valyaev, Frontend Engineer, and Travis Turner, Tech Editor

Topics: iOS, JavaScript, CSS, TypeScript

Read how to accurately detect Safari and iOS versions using WebKit feature checks, behavioral tests, and selective UA hints to gate features safely and avoid breaking UX.

Why is accurately detecting the version of Safari and iOS you're dealing with so important for modern web development? The reasons are seriously many: applying fixes/enhancements only where needed, preventing confusion for users on other browsers, displaying the right prompts for actions or installations, enabling or disabling features, providing accurate analytics, and support users with tailored instructions, especially since some browsers “masquerade” in their user agent strings or behave differently in WebViews. In this post, we'll talk about how to identify Safari/iOS versions with ease in 2025 (and beyond) .

[Read more](#)

A Knock at the Door

Colin Vanderburg · n+1 Magazine

No one is happy with NASA's new idea for private space stations

Eric Berger · Ars Technica

Most elements of a major NASA event this week that laid out spaceflight plans for the coming decade were well received: a Moon base, a focus on less talk and more action, and working with industry to streamline regulations so increased innovation can propel the United States further into space.

However, one aspect of this event, named Ignition, has begun to run into serious turbulence. It involves NASA's attempt to navigate a difficult issue with no clear solution: finding a commercial replacement for the aging International Space Station.

During the Ignition event on Tuesday, NASA leaders had blunt words for the future of commercial activity in low-Earth orbit. Essentially, they are not confident in the viability of a commercial marketplace for humans there, and the agency's plan to work with private companies to develop independent space stations does not appear to be headed toward success. Plenty of people in the industry share these concerns, but NASA officials have not expressed them out loud before.

[Read full article](#)

Optimistic UI in Rails with optimism... and Inertia

Travis Turner · Evil Martians Chronicles

Authors: Svyatoslav Kryukov, Backend Engineer, and Travis Turner, Tech Editor

Topics: Rails, React, JavaScript

Build optimistic UI in Rails with Inertia by updating props before requests complete. Learn the replaceProp pattern, automatic reconciliation, rollback behavior, and history caveats using a kanban board example.

Your user drags a card across a kanban board. They expect instant feedback. Not a loading spinner, brief flicker, or “please wait.” Here's the thing: optimistic UI is a lie. And the modern web has trained everyone to expect interfaces that lie convincingly. We show the user what we expect to happen before the server confirms it. With Inertia Rails, telling that lie takes remarkably little code because of how Inertia handles state.

[Read more](#)

Tutorialkit.rb: interactive Ruby tutorials entirely in the browser

Travis Turner · Evil Martians Chronicles

I think of the people I met on canvasses. The older Polish woman in Greenpoint who took a thick stack of Zohran flyers to give out to all her friends. The hijabi Indian American mother and daughter who drove in from Long Island to knock doors for Zohran so that, the mother said, life could be as affordable for others as it was when she was growing up in the Bronx. A mobility-impaired man in Bay Ridge who said he rarely got visitors and invited me into his apartment, where he talked about his frustration with inaccessible transit, and the hope Zohran's platform held for him.

Authors: Albert Pazderin, Backend Engineer, Vladimir Dementyev, Principal Backend Engineer, and Travis Turner, Tech Editor

Topics: Open Source, Rails, Local-first, DX, WebAssembly, Ruby, JavaScript

The final report for Ruby Association Grant on TutorialKit.rb—a toolkit for building interactive Ruby and Rails tutorials that run entirely in the browser using WebAssembly and WebContainers. Featuring a full-featured installer, agent-friendly development workflow, deployment pipelines, HTTP support, and real-world examples.

TutorialKit.rb, a toolkit for building interactive Ruby and Rails tutorials that run entirely in the browser, has reached the release candidate stage and is ready for general use. In this post, we want to quickly walk through the final DX for building tutorials, share some real-world examples, and the research behind the project.

[Read more](#)

Listen to a grieving mother and have no doubts: water privatisation has been a lethal scandal | Clive Lewis

Clive Lewis · The Guardian Environment

Progressives must wage a battle against a rotten capitalist system that milks our resources – and destroys lives. Let the fightback start with water

In more than a decade as an MP, I have attended hundreds of meetings in parliament. Most pass. Some linger. Few stay with you. But a recent event was very different.

We hosted the actors, the real-life people they portrayed and the production team behind the Channel 4 docudrama *Dirty Business*. It tells the story of campaigners and families who have spent years fighting not just privatised water companies, but a system that was meant to protect them – and has too often failed.

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ANALYSIS

The post [How Panic Lost Me Money In A Non-Investment Way](#) appeared first on [Financial Samurai](#) .

Chelsea Kirk, n+1 Magazine: What scares them is something else: the realization spreading across Los Angeles that the private housing market isn't just failing wildfire victims—it's failing by design. That the inability to meet this moment isn't the result of a few bad landlords, it's a feature of a system built to extract. What landlords fear is that we might imagine something better: a world where housing isn't a commodity at all, a world without landlords.

Rachel Khong, n+1 Magazine: “I only love swimming,” I said. And it was true that I’d always felt awkward on land. Sometimes I wondered if that was why I was so unsuccessful at online dating: technology was incompatible with the water. “The monk thing does appeal.”

Olivia Paschal, n+1 Magazine: In retrospect, the week of Donald Trump's inauguration was an inopportune time for Walmart to hold the grand opening of its new corporate campus.

Mark Greif, n+1 Magazine: What is the function of demanding that the entire federal bureaucracy search itself for traces that it might be funding programs to which the Leader might object? Plainly, to dement it. Or make it paranoid, make it pusillanimous, make it humiliated, and bring to the forefront the narcs, toadies, informers, and eager collaborators who can be promoted to replace those who continue their jobs and assert the law.

Helena Horton, The Guardian Environment: Exclusive: critics warn Reform UK use of trade policy would increase food costs amid cost-of-living crisis

Arable farmer and campaigner Clive Bailye has been appointed as a farming and land use adviser for Reform UK. Bailye owns the website The Farming Forum, a social network for farmers, and helped organise the large-scale protests against the Labour government's introduction of inheritance tax for farmed land.

Andrea More, n+1 Magazine: That night I dreamed that my dad had died. The lectern at the service was too high and nobody could see me. I didn't know what to say, so I gave a eulogy about someone I knew a little better who was still alive. When I woke up, my pillow was drenched with sweat. Had my speech been well received? I did some googling and learned that lecterns are between forty-five and forty-eight inches tall. I am nearly a foot taller than that. Thank God.

Lydia Pelot-Hobbs, David Stein, n+1 Magazine: Countless people are left in harm's way because of the state's refusal to make evacuation plans. The hurricane ends up coming through weaker than expected—but then the levees break. Within hours, 80 percent of the city is underwater. ~~The hurricane strikes the Gulf Coast of Louisiana, where it becomes known as the Gulf of Mexico hurricane.~~

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